Relic Neutrinos

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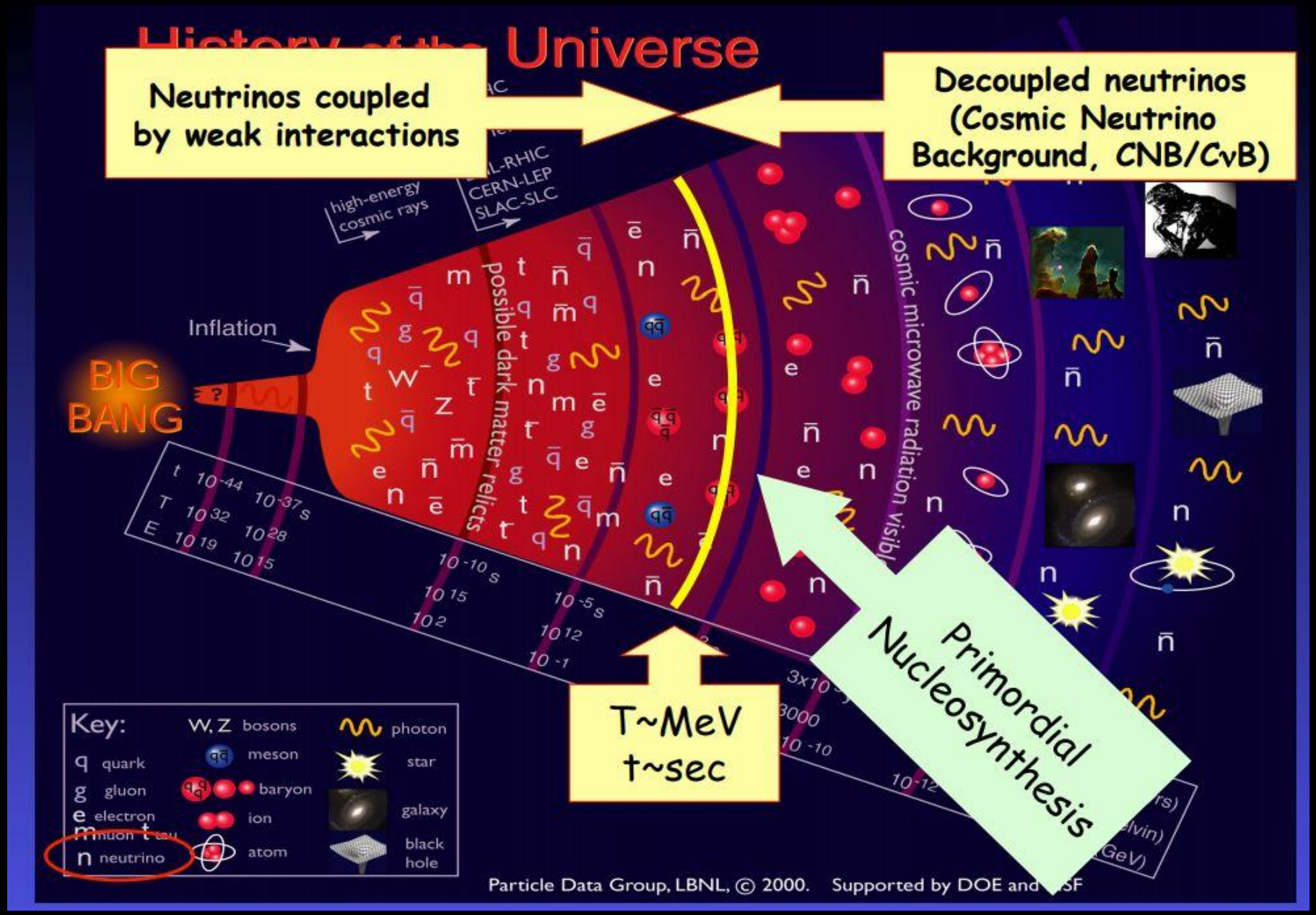

Sergio Pastor, Neutrino astrophysics and cosmology, Nu TheME Workshop CERN, September 2010

Neutrino Decoupling

In early Universe:

$$\upsilon + \bar{\upsilon} \leftrightarrow e^{+} + e^{-},$$
$$\upsilon + e \leftrightarrow \upsilon + e^{+}$$

Neutrinos decouple when the expansion rate exceeds the interaction rate

Neutrino Decoupling

Interaction rate:

$$\Gamma = n < \sigma v >$$

where n is the number density of target particles, σ is the cross-section and $v \approx 1$ is the neutrino velocity.

For weak interaction:

$$< \sigma v > \sim G_F^2 \, T_\gamma^2$$

& the number density of relativistic particles:

$$n \sim T_\gamma^3$$

So:

$$\Gamma \sim G_F^2 \, T_\gamma^5$$

Neutrino Decoupling

For a radiation-dominated Universe:

$$H \sim \frac{T_\gamma^2}{M_P}$$

$$\Gamma \sim H \rightarrow T_\gamma \sim 1\, Mev$$

Photon Temperature

- In thermal equilibrium the entropy S is conserved. so it's useful to define entropy density $$s = \frac{S}{V}$$
- Conservation of entropy in the expanding Universe implies
$$s \propto R^{-3}$$
- Another hand we can write this function of T_γ $$s = \frac{2\pi^2}{45} g_s T_\gamma^3$$
- Thus $T_\gamma \propto g_s^{-1/3} R^{-1}$ and more precise calculations
$$T_\gamma = T_\gamma^0 \left(\frac{4}{11}\right)^{\frac{1}{3}} (1+z)$$

Neutrino Temperature

- Since neutrinos were relativistic at decoupling $T_\nu \propto R^{-1}$
- Another word we can write $T_\nu = T_\nu^0\,(1+z)$
- By compairing with photon temperature $T_\nu^0 = \left(\frac{4}{11}\right)^{\frac{1}{3}} T_\gamma^0$
- Since both the neutrino and photon temperatures scaled as R^{-1} after $e^+ - e^-$ annihilation $T_\nu = \left(\frac{4}{11}\right)^{\frac{1}{3}} T_\gamma$
- From the measured temperature of the CMB $T_\nu^0 = (1.676 \pm 0.001) \times 10^{-4}$ eV

Neutrino Detection

- GRAVITATIONAL CLUSTERING OF RELIC NEUTRINOS
- DETECTION BY COHERENT SCATTERING
- DETECTION USING HIGHLY ENERGETIC COSMIC NEUTRINOS
- DETECTION USING NEUTRINO CAPTURE ON UNSTABLE NUCLEI

References

- Carlo Giunti, Chung W. Kim - Fundamentals of Neutrino Physics and Astrophysics-Oxford University Press, USA (2007)
- Petr Vogel, How difficult it would be to detect Cosmic Neutrino Background?, https://doi.org/10.1063/1.4915587
- Valerie Domcke and Martin Spinrath, Detection prospects for the Cosmic Neutrino Background using laser interferometers, arXiv:1703.08629v2
- Sergio Pastor, Relic Neutrino: Neutrino properties from cosmology, arXiv:0306233v1

Thank You!